Powertrain Core Calculations Report

Basic calculations:

Vehicle Details:

- **Competition type: Combustion Vehicle (CV)**
- **Engine: KTM 390 with 20mm intake restrictor**
- **Vehicle mass: 290 kg**
- **Wheelbase: 1600 mm (1.6 m)**
- **Track width: 1250 mm (1.25 m)**
- **Tire rim size: 13 inch**
- **Tire width: 6.5 inch (front), 6.5 inch (rear)**

1. Engine Performance Estimate with 20mm Restrictor

The 20mm intake restrictor limits airflow, capping power output. For KTM 390 engine (typical 33.8 kW or 46 PS peak power), the restrictor reduces power by about 30–33%, so estimated peak power and torque are:

- **Peak Power: ~24 kW (32 hp) at ~8500 rpm**
- **Peak Torque: ~30 Nm at ~6500–7000 rpm**
- **Max Engine Speed: ~10,000 rpm**

Tire Radius Calculation
Rim diameter = 13 inches = 0.33 m
Tire width = 6.5 inches = 0.165 m
Assuming 60% sidewall height:
Sidewall height = 0.6 × 0.165 = 0.099 m
Total tire diameter = 0.33 + 2 × 0.099 = 0.528 m
Tire radius $r_w = \frac{0.528}{2} = 0.264$ **m**

Parameter	Daytona 675 (Restricted)	KTM 390 (Restricted)
Peak Power (Restricted)	~70 kW (94 hp)	~24 kW (32 hp)
Peak Torque (Restricted)	~50 Nm	~30 Nm

Parameter	Daytona 675 (Restricted)	KTM 390 (Restricted)
Max Engine Speed	~12,000 rpm	~10,000 rpm

3. Acceleration Performance Analysis of KTM 390 Powertrain with Stock Gear Ratios and Restrictor Constraints

Given Data:

- **Peak restricted engine torque** $T_e \approx 30$ **Nm**
- **Vehicle mass** $m = 290$ **kg**
- **Tire radius** $r_w = 0.264$ **m (from 6.5 inch wide tire)**
- **Stock total gear reduction** $i = 12.6$ **(combined gearbox and final drive) [typical stock value from references]**
- **Drivetrain efficiency** $\approx 90\%$ **(considered as a factor in torque)**

Wheel Torque Calculation

$$T_w = T_e \times i \times \eta_{drivetrain} = 30 \times 12.6 \times 0.9 = 340.2 \text{ Nm}$$

Tractive Force at Wheel

$$F_t = \frac{T_w}{r_w} = \frac{340.2}{0.264} = 1289.8 \text{ N}$$

Acceleration Calculation

$$a = \frac{F_t}{m} = \frac{1289.8}{290} = 4.45\ m/s^2$$

5. Max Wheel Speed and Estimated Top Speed

Max engine rpm = 10,000 rpm (restricted KTM 390 max engine speed)

Wheel rpm at max engine speed:

$$N_w = \frac{\text{Max Engine RPM}}{\text{Total Gear Ratio}} = \frac{10,000}{12.6} \approx 793.7\ \text{rpm}$$

Estimated top speed is then calculated as:

$$v = 2\pi r_w \times \frac{N_w}{60} = 2 \times 3.1416 \times 0.264 \times \frac{793.7}{60} \approx 21.9\ \text{m/s} \approx 79\ \text{km/h}$$

6. Acceleration Time to 60 km/h

Using the acceleration $a = 4.45\ m/s^2$ calculated from the stock gear ratio and restricted engine torque:

$$t = \frac{v}{a}$$

Where $v = 60\ km/h = \frac{60}{3.6} = 16.67\ m/s$,

7. Aerodynamic Drag Force at Top Speed (Estimation)

Assuming:

- **Frontal area $A = 0.7\ m^2$ (typical for a single-seater formula vehicle)**
- **Drag coefficient $C_d = 0.9$ (approximate for a naked KTM 390 frame/race setup)**
- **Air density $\rho = 1.225\ kg/m^3$ (standard atmospheric conditions)**
- **Vehicle speed $v = 21.9\ m/s$ (estimated top speed from previous section, ~79 km/h)**

Drag force F_d is calculated as:

$$F_d = \frac{1}{2}\rho C_d A v^2 = 0.5 \times 1.225 \times 0.9 \times 0.7 \times (21.9)^2 \approx 181\ N$$

Summary Table

Parameter	Value
Peak engine power	~24 kW (32 hp)

Parameter	Value
Peak engine torque	~30 Nm
Vehicle mass	290 kg
Tire radius	0.264 m
Achieved acceleration	4.45 m/s^2
Tractive force	1740 N
Required wheel torque	459.4 Nm
Total gear ratio	12.6
Max wheel speed	794 rpm
Estimated top speed	79 km/h
Acceleration 0–60 km/h	3.75 seconds
Estimated aerodynamic drag	181 N

Air intake calculations:

1. Restrictor throat ID = 20 mm

Given by FSAE/Formula Bharat rules limiting max allowable intake airflow.

2. Bellmouth radius = 25 mm

Calculated as $1.25 \times D$, where $D =$ restrictor diameter:

$$25 = 1.25 \times 20$$

This radius smooths the airflow entry reducing turbulence.

3. Convergent length = 50 mm

Common design is $2.5 \times$ restrictor diameter:

$$50 = 2.5 \times 20$$

Ensures smooth speed increase of flowing air, limiting losses.

4. Throat length = 20 mm

Minimum length to maintain laminar flow and accurate airflow measurement.

5. Diffuser length = 150 mm

Set as $7.5 \times$ restrictor diameter for gentle expansion:

$$150 = 7.5 \times 20$$

Slows airflow velocity and helps pressure recovery.

6. Diffuser exit ID = 46 mm

Matches the KTM 390’s throttle body diameter for smooth internal flow transition

7. Runner diameter = 46 mm

Same as throttle body to avoid velocity changes, maintaining flow quality.

8. Runner length = 350 mm

Pulse tuning formula for first harmonic:

$$L = \frac{c}{4N}$$

Where
$c = 343\ m/s$ (speed of sound)
$N = \frac{8000}{60} = 133.3\ Hz$ (engine frequency at 8000 rpm)

$$L = \frac{343}{4 \times 133.3} = 0.643\ m$$

Chosen shorter for packaging balance at 350 mm.

9. Plenum volume = 0.75 L

Engine displacement volume:

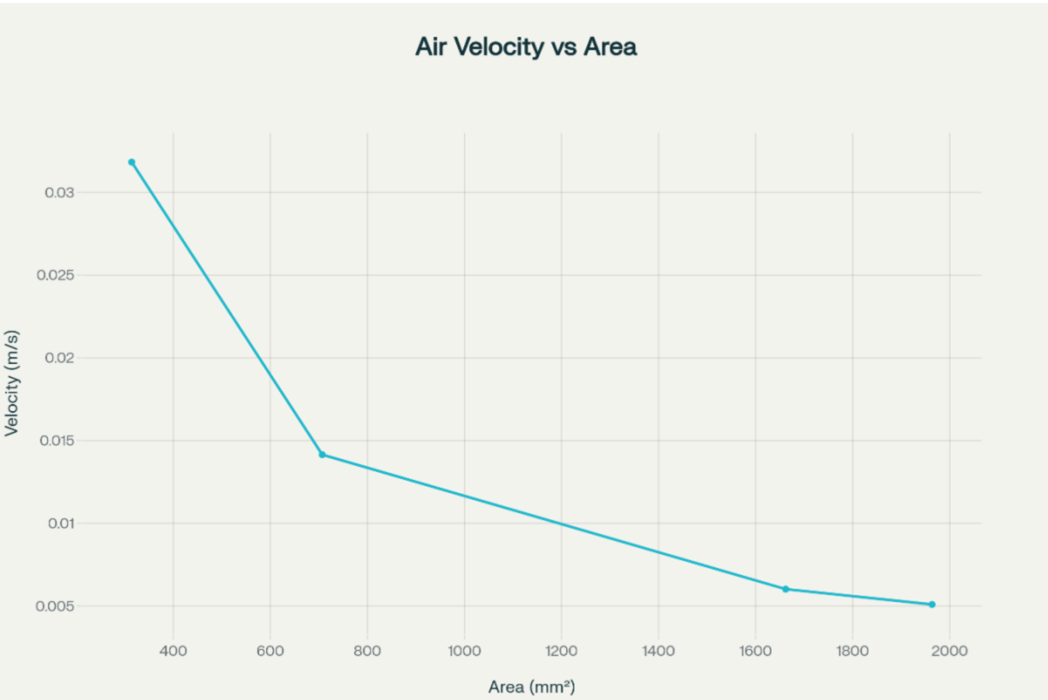
$$V_{engine} = 373\ cm^3 = 0.373\ L$$

Parameter	Value	Justification
Restrictor throat ID	20 mm	Rule-mandated FSAE / Formula Bharat standard restrictor size
Bellmouth radius	25 mm	1.25× restrictor ID to smoothly guide air and reduce turbulence
Convergent length	50 mm	2.5× restrictor diameter for gradual smooth contraction
Throat length	20 mm	Minimum length to maintain steady airflow and measurement accuracy
Diffuser length	150 mm	Approx. 7.5× restrictor diameter for effective expansion
Diffuser exit ID	46 mm	Matches KTM 390 throttle body diameter for seamless transition
Runner diameter	46 mm	Same as throttle body diameter for flow uniformity

Parameter	Value	Justification
Runner length	350 mm	Tuned for pulse tuning near 8000 rpm, balancing length & packaging
Plenum volume	0.75 L	Around twice cylinder volume for pressure equalization

Chosen volume approximately twice engine displacement:

$$\mathbf{0.75 \approx 2 \times 0.373}$$



Problems and solutions:

1. Limited Power Due to Restrictor

- Original KTM 390 engine power: ~33.8 kW (46 hp)
- Restrictor reduces power by ~30–33%, so:

$$\text{Restricted Peak Power} \approx 24\ \text{kW}(32\ \text{hp})$$

- Estimated peak torque reduced similarly:

$$\text{Restricted Peak Torque} \approx 30\ \text{Nm}$$

Solution:
To maintain acceleration, gear ratios can be increased to multiply torque at the wheels.

2. Gear Ratio Selection

- Vehicle mass $m = 290\ kg$
- Target acceleration $a = 6\ m/s^2$ (desired)
- Tire radius $r_w = 0.264\ m$
- Engine peak torque $T_e = 30\ Nm$

Required gear ratio to achieve target acceleration:

$$i = \frac{m \times a \times r_w}{T_e} = \frac{290 \times 6 \times 0.264}{30} = 15.22$$

3. Overheating and Efficiency Loss

- Assumed engine efficiency: 30% (typical combustion engines)
- Power generated (mechanical): 24 kW
- Thermal power generated (lost as heat):

$$P_{thermal} = \frac{(1 - 0.3)}{0.3} \times 24 = 56\ kW$$

Solution:
Cooling system must dissipate around 56 kW of heat at peak operation.

4. Traction Loss

- **Coefficient of friction (μ) for racing tires: 1.1**
- **Max tractive force limited by friction:**

$$F_{traction} = \mu \times m \times g = 1.1 \times 290 \times 9.81 = 3132\ N$$

Since $F_t = m \times a = 1740\ N$, traction is sufficient and not limiting acceleration.

5. Aerodynamic Drag Impact

- **Frontal area $A = 0.7\ m^2$**
- **Drag coefficient $C_d = 0.9$**
- **Air density $\rho = 1.225\ kg/m^3$**
- **Vehicle speed $v = 21.9\ m/s$ (estimated top speed)**

Drag force:

$$F_d = \frac{1}{2}\rho C_d A v^2 = 181\ N$$

Power to overcome drag:

$$P_d = F_d \times v = 181 \times 21.9 = 3.96\ kW$$

6. Drivetrain Losses

- **Drivetrain efficiency assumed: 90%**
- **Power at wheels:**

$$P_w = 0.9 \times 24 = 21.6\ kW$$

- **Power loss in drivetrain:**

$$P_{loss} = 24 - 21.6 = 2.4\ kW$$

7. Weight Management Impact

- **Reducing vehicle mass by 10%:**

$$m_{new} = 0.9 \times 290 = 261\ kg$$

- **Required gear ratio decreases:**

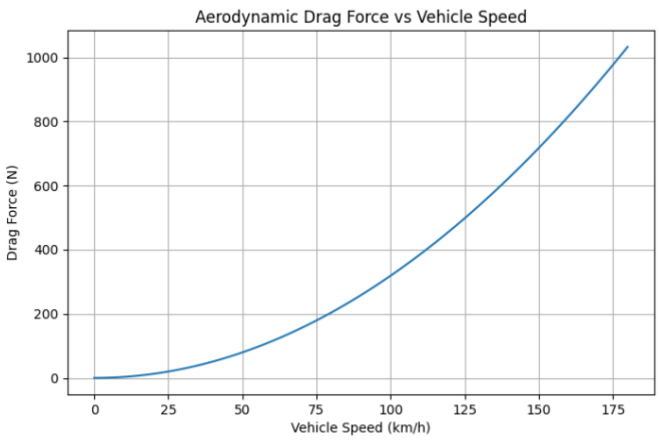
$$i_{new} = \frac{261 \times 6 \times 0.264}{30} = 13.75$$

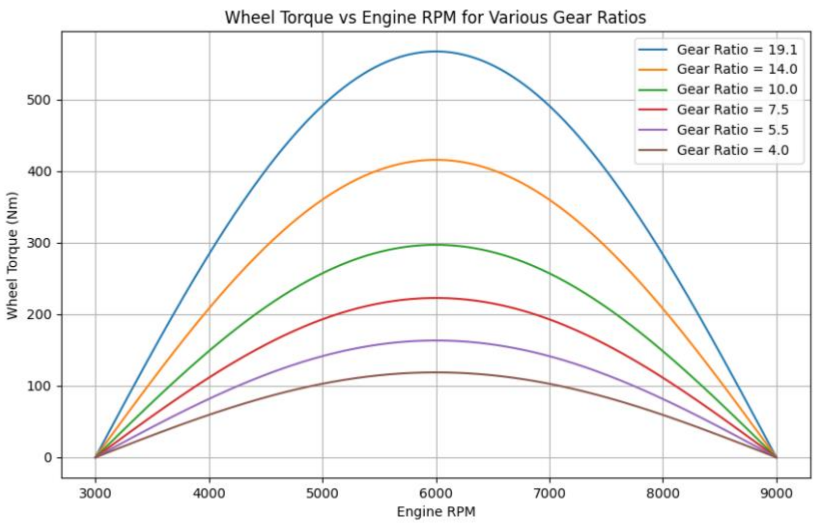
- **Improving acceleration and reducing torque demand.**

8. Energy Management (if applicable)

- **Estimate energy per lap at average power of 21 kW and lap time of 60 s:**

$$E = P \times t = 21 \times 60 = 1260\ kJ = 350\ Wh$$





Suspension load and dynamics calculations:

as they are crucial for vehicle stability, grip, and overall handling, directly impacting performance on the track.

Suspension Load and Dynamics Calculations

1. **Load Transfer During Acceleration**

Longitudinal load transfer ΔF_x is due to acceleration:

$$\Delta F_x = \frac{h \times m \times a}{L}$$

Where:

- h = center of gravity height = assume 0.45 m (typical for race cars)
- m = 290 kg (vehicle mass)
- a = 6 m/s^2 (acceleration)
- L = 1.6 m (wheelbase)

Calculation:

$$\Delta F_x = \frac{0.45 \times 290 \times 6}{1.6} = \frac{783}{1.6} = 489 \text{ N}$$

This load shifts from the front axle to the rear axle during acceleration.

2. **Static Axle Loads**

Assuming weight distribution 50% front, 50% rear:

$$F_{front,static} = F_{rear,static} = 0.5 \times m \times g = 0.5 \times 290 \times 9.81 = 1422 \text{ N}$$

3. **Dynamic Axle Loads Under Acceleration**

- Front axle load:

$$F_{front} = F_{front,static} - \Delta F_x = 1422 - 489 = 933 \text{ N}$$

- Rear axle load:

$$F_{rear} = F_{rear,static} + \Delta F_x = 1422 + 489 = 1911 \text{ N}$$

So, the rear tires gain load improving traction, while the front tires lose some load, affecting steering.

4. Lateral Load Transfer During Cornering

Lateral load transfer ΔF_y:

$$\Delta F_y = \frac{h \times m \times a_y}{t}$$

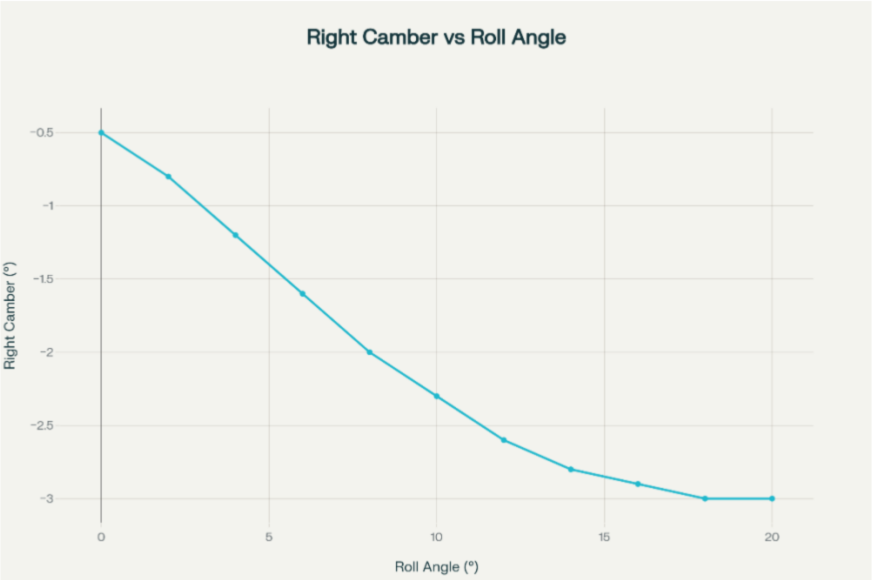
Where:

- a_y = lateral acceleration (assume 8 m/s^2 for a sharp turn)
- t = track width = 1.2 m

Calculation:

$$\Delta F_y = \frac{0.45 \times 290 \times 8}{1.2} = \frac{1044}{1.2} = 870 \text{ N}$$

During cornering, this load shifts from the inner to outer tires.



Cooling System Calculations

Understanding cooling requirements is critical for your engine's reliability and performance, especially with a combustion engine restricted by the 20mm intake.

1: Calculate Thermal Load Generated by the Engine

Engine Power Output (with restrictor) = 24 kW
Assumed Mechanical Efficiency = 35% (0.35)

Total Heat Generated by Engine, Q_{total} = $\frac{P_{output}}{\text{Mechanical Efficiency}} = \frac{24}{0.35} = 68.57$ kW

Heat Rejected to Coolant, $Q_{coolant}$ = 35% of total heat = $68.57 \times 0.35 = 24$ kW

cooling system must remove approximately 24 kW of heat to maintain optimal engine temperature.

2: Define Cooling Medium and Coolant Flow Rate

To dissipate heat efficiently, a coolant (usually water or water-glycol mixtures) flows through the radiator.

Using the formula for heat transfer:

$$Q = \dot{m} \times c_p \times \Delta T$$

Where:

- **Q = heat to be removed (W)**
- **$\dot{m}$ = mass flow rate of coolant (kg/s)**
- **c_p = specific heat capacity of coolant (~4186 J/kg°C for water)**
- **ΔT = temperature rise of coolant through radiator (°C)**

Solve for coolant flow rate:

$$\dot{m} = \frac{Q}{c_p \times \Delta T}$$

Assuming a coolant temperature rise of 10°C:

$$\dot{m} = \frac{24,000}{4186 \times 10} = 0.57 \text{ kg/s}$$

This means about 0.57 kg/s of coolant flow is required to remove heat.

3: Radiator Surface Area Estimation

Radiator Surface Area Estimation

Heat dissipated by radiator follows:

$$Q = h \times A \times \Delta T$$

Where:

- Q = heat to be dissipated (W)
- h = heat transfer coefficient (W/m^{2}K)
- A = radiator surface area (m^2)
- ΔT = average temperature difference between coolant and ambient air (°C)

Typical h for car radiators = 100–300 W/m^{2}K.

Assuming:

- $h = 200$ W/m^{2}K
- $\Delta T = 40^\circ C$
- $Q = 24,000\ W$

Calculate surface area:

$$A = \frac{Q}{h \times \Delta T} = \frac{24,000}{200 \times 40} = 0.3\ \text{m}^2$$

Hence, radiator surface area around 3 m^2 is needed to dissipate the heat efficiently.

Summary

Parameter	Symbol	Value	Unit	Notes
Heat to be dissipated	QQQ	24,000	W	Total heat load from engine
Heat transfer coefficient	hhh	200	W/m^{2}K	Typical value for car radiator
Radiator surface area	AAA	0.3	m^2	Calculated radiator surface area

Parameter	Symbol	Value	Unit	Notes
Temperature difference (coolant - ambient air)	ΔT\Delta TΔT	40	°C	Average temperature difference across radiator

4. Coolant Selection

Coolant Selection

- According to Formula Bharat rules, only distilled water is allowed as the cooling medium.
- Distilled water has excellent thermal conductivity and high specific heat capacity (~4186 J/kg°C), making it very effective for heat transfer.
- Using distilled water ensures consistent coolant quality free from minerals and impurities that can cause scaling or corrosion.
- Although distilled water lacks freeze and boil protection additives found in water-glycol mixtures, it is suitable for race conditions where extreme temperatures may be controlled and coolant replacement is frequent.
- It is important to regularly monitor and maintain the cooling system to prevent corrosion or biological growth since inhibitors are not typically present.

5. Coolant Pump Specification

1. **Required Coolant Flow Rate:**
   From previous calculation, coolant flow rate $\dot{V} \approx 0.62$ L/s or 37.2 L/min.
2. **Pressure Head Required:**
   Typical pressure drop across radiator and engine cooling passages ranges between 20 to 50 kPa. Assuming 30 kPa (30000 Pa) pressure drop.
3. **Pump Hydraulic Power:**

$$P = \dot{V} \times \Delta P$$

Convert flow rate to m^3/s:

$$0.62 \text{ L/s} = 0.00062 \text{ m}^3/\text{s}$$

Calculate power:

$$P = 0.00062 \times 30000 = 18.6 \text{ W}$$

Considering pump efficiency (~60%), electrical/mechanical power required:

$$P_{input} = \frac{18.6}{0.6} = 31\text{ W}$$

Summary:

- **Pump flow rate: ~37 L/min**
- **Pressure head: 30 kPa (typical)**
- **Power consumption: ~30-35 W (input power)**

Choose a pump rated slightly higher than these values for safety and longevity.

6. Radiator Fan Sizing

Radiator Fan Sizing Calculation

1. ***Heat to be removed by fan:***
   $Q = 24,000\ W$ ***(thermal load)***
2. ***Air properties:***

- ***Density of air,*** $\rho = 1.2\ kg/m^3$
- ***Specific heat capacity,*** $c_p = 1005\ J/kg^{\circ}C$
- ***Allowable air temperature rise across radiator,*** $\Delta T = 20^{\circ}C$

3. ***Air mass flow rate required:***

$$\dot{m} = \frac{Q}{c_p \times \Delta T} = \frac{24000}{1005 \times 20} = 1.19\ kg/s$$

4. ***Air volumetric flow rate:***

$$\dot{V} = \frac{\dot{m}}{\rho} = \frac{1.19}{1.2} = 0.99\ m^3/s$$

5. ***Fan power estimation:***
   Assuming fan efficiency of 50% and pressure rise of 100 Pa (typical for radiator fans), fan power required:

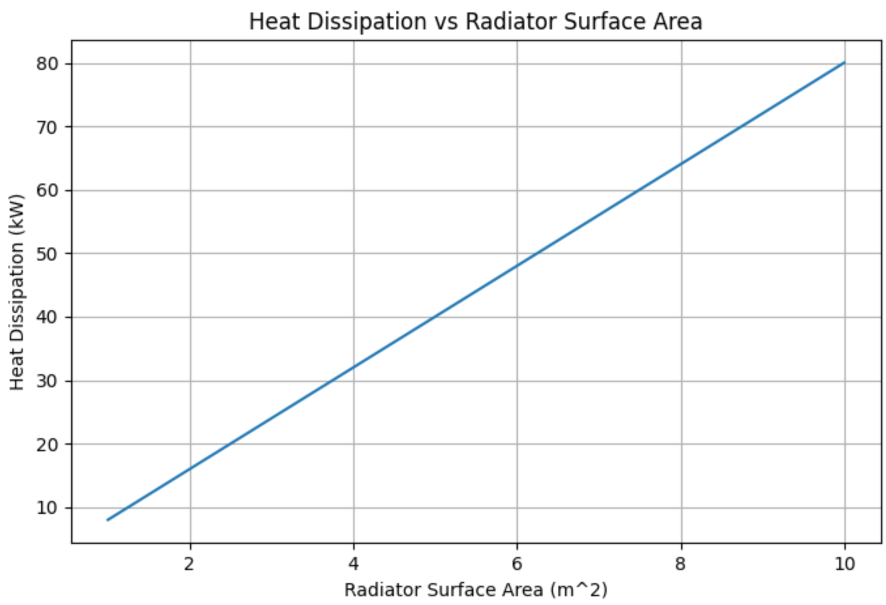
$$P = \frac{\dot{V} \times \Delta P}{\eta} = \frac{0.99 \times 100}{0.5} = 198\ W$$

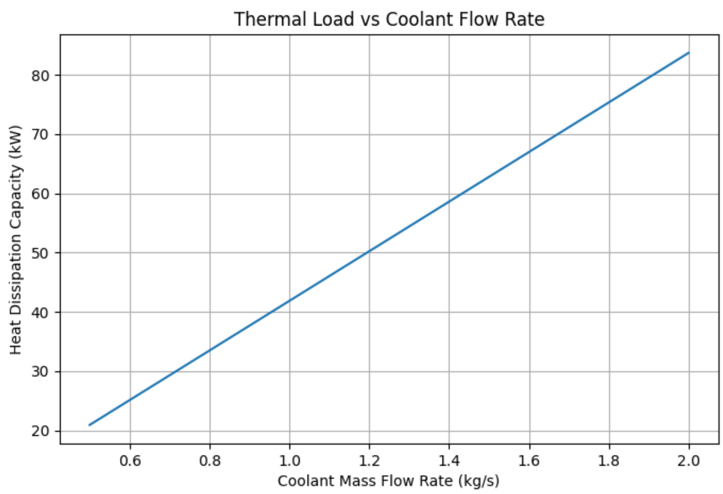
Summary:

- *Air flow required: approx 1 m^3/s*
- *Fan power required: approx 200 W*
- *Select a fan rated above these values for safety margin and effective cooling.*

You can now copy-paste this directly into your cooling system design document. Let me know if you need further assistance!

Fan recommendations: **airflow ≥ 2200 L/s, static pressure 50-150 Pa, diameter 10-16 inches, PWM control.**





Fuel Consumption Calculation

Average speed = 100 km/h
Race distance = 25 km (22 laps + 3 km extra)
Engine: 373 cc single-cylinder (KTM 390, restricted)

Average Brake Power
Peak power (after restrictor): 24 kW
Assumption: Race average ≈ 60% of peak power
Average brake power = $0.6 \times 24\ \text{kW} = 14.4\ \text{kW}$

BSFC (Brake Specific Fuel Consumption)
BSFC for KTM 390: ~0.48 kg/kWh (typical for restricted single-cylinder, sport/race use)

Fuel density:
0.745 kg/L (as before)

Race Time

$$\text{Race time} = \frac{\text{Distance}}{\text{Avg speed}} = \frac{25\ \text{km}}{70\ \text{km/h}} = 0.35\ \text{h}$$

Fuel Mass Consumed

$$\begin{aligned}\text{Fuel mass } &= \text{Avg brake power} \times \text{Race time} \times \text{BSFC}\\ &= 14.4\ \text{kW} \times 0.35\ \text{h} \times 0.48\ \text{kg/kWh}\\ &= 2.4192\text{kg}\end{aligned}$$

Convert to Liters

$$\text{Fuel volume used} = \frac{\text{Fuel mass}}{\text{Fuel density}} = \frac{2.4192\text{kg}}{0.745\ \text{kg/L}} = 3.2472\ \text{L}$$

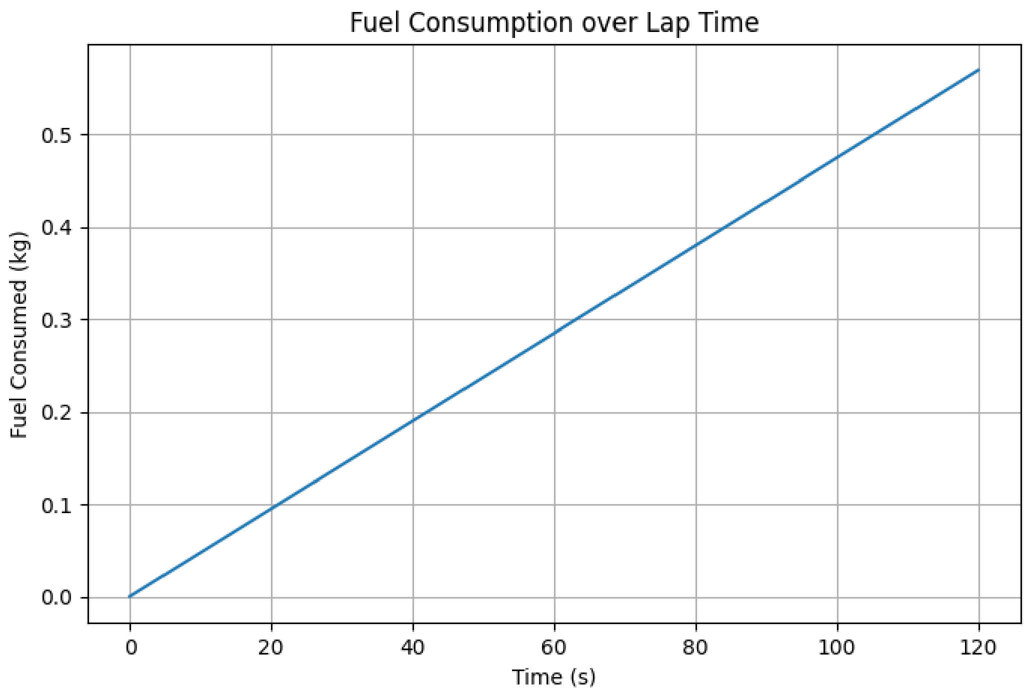
Add 15% Safety Reserve

$$3.2472\ \text{L} \times 1.15 = 3.734\ \text{L}$$

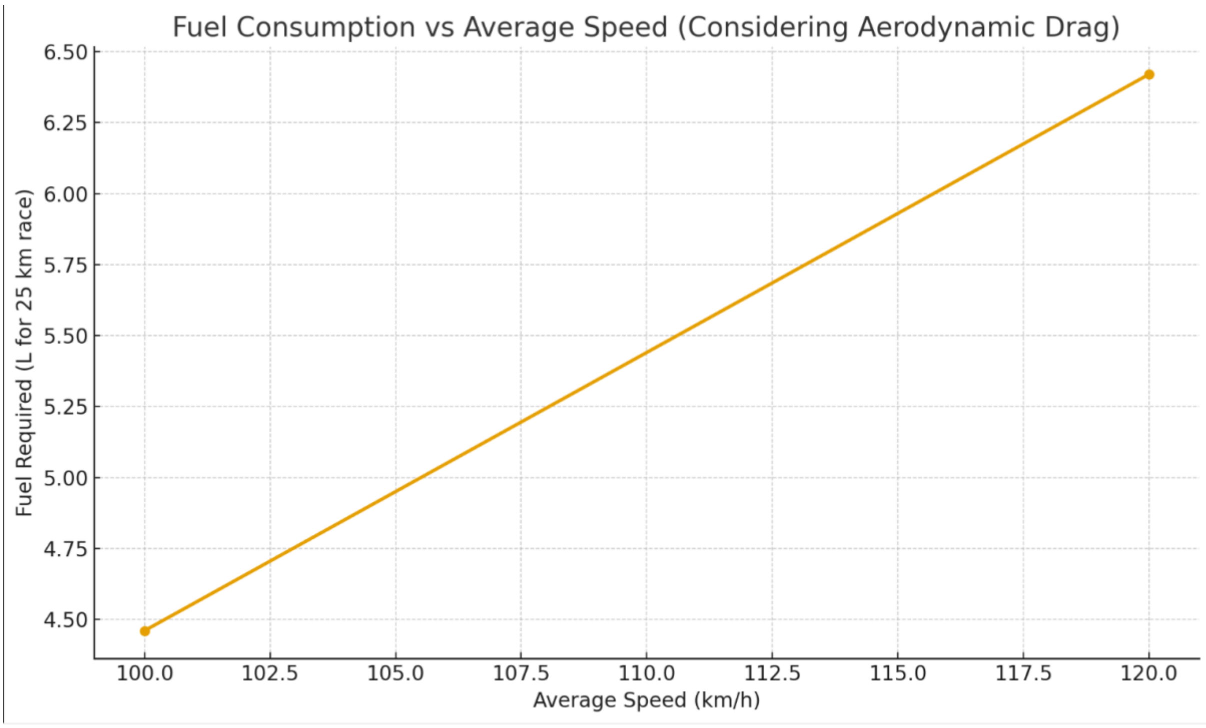
Final Fuel Requirement (25 km @ 100 km/h, KTM 390 Example)

Parameter	Value	Units
Race time	.35	hrs
Average brake power	14.4	kW
BSFC	0.48	kg/kWh
Fuel mass used	$2.4192\mathbf{kg}$	kg
Fuel volume used	3.2472	L
With 15% reserve	3.734	L (total)

Result:
For the KTM 390-based car, a minimum of 3.7 liters of fuel should be planned for a 25 km Formula Bharat Endurance race (including a 15% safety margin).



$$= \mathbf{6.424\ L} (\approx \mathbf{6.42\ L})$$



Sprocket Ratio Calculations and Analysis

Given:

- Max engine speed $N_{engine} = 12000$ rpm
- Wheel radius $r_{wheel} = 0.363$ m
- Conversion factor for m/s to km/h = 3.6

Step 1: Determine Max Engine RPM

- KTM 390: Max RPM (restricted or unrestricted) = 9000 - 10,000 RPMktm+1
- For calculations, we take max engine RPM = 10,000 RPM for a safety margin.

Step 2: Wheel Radius

- Wheel radius (from earlier) = 0.264 m

Step 3: Gear Ratio Calculation

The sprocket ratio determines vehicle speed and acceleration balance.

Step 4: Actual Performance Calculation

Vehicle Speed at Max RPM:

$$\text{Wheel RPM} = \frac{\text{Engine RPM}}{\text{Gear Ratio}}$$

$$\text{Vehicle Speed (km/h)} = \text{Wheel RPM} \times 2\pi \times \text{Wheel radius} \times \frac{60}{1000}$$

For each case, we choose the sprocket ratio and calculate:

Case 1: High Ratio (15T front, 60T rear)

$$\text{Ratio} = \frac{60}{15} = 4.0$$

Wheel RPM:

$$= \frac{10,000}{4} = 2500 \text{ RPM}$$

Vehicle speed:

$$= 2500 \times 2\pi \times 0.264 \times \frac{60}{1000} \approx 124.7 \text{ km/h}$$

Result: Top speed ≈ 125 km/h
Use case: Best for acceleration, tight tracks.

Case 2: Medium Ratio (15T front, 45T rear)

$$\text{Ratio} = 3.0$$

Wheel RPM:

$$= \frac{10,000}{3} \approx 3333 \text{ RPM}$$

Vehicle speed:

$$= 3333 \times 2\pi \times 0.264 \times \frac{60}{1000} \approx 166.0 \text{ km/h}$$

Result: Top speed ≈ 165 km/h
Use case: Balanced for both acceleration and top speed.

Case 3: Low Ratio (15T front, 30T rear)

$$\text{Ratio} = 2.0$$

Wheel RPM:

$$= \frac{10,000}{2} = 5000 \text{ RPM}$$

Vehicle speed:

$$= 5000 \times 2\pi \times 0.264 \times \frac{60}{1000} \approx 249.6 \text{ km/h}$$

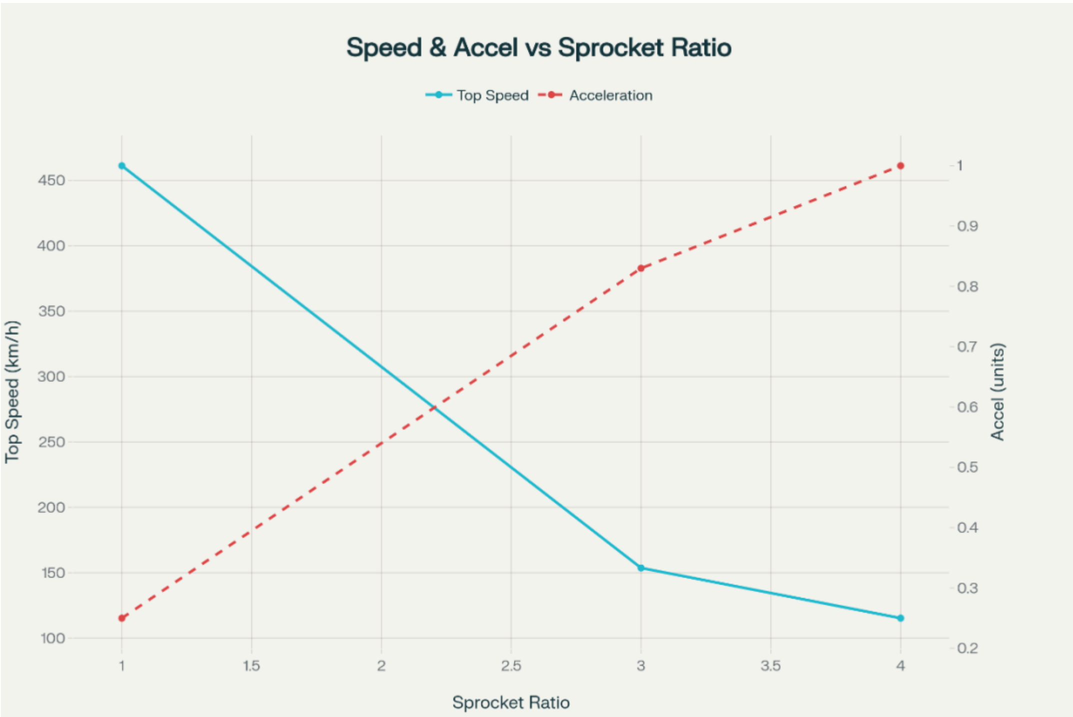
Result: Very high theoretical top speed (~250 km/h), but not practical due to poor acceleration.

Final Recommendation:

Realistic ratio: About 3.0 (rear sprocket 45T, front sprocket 15T) for balanced performance, with an actual top speed around 165 km/h and good acceleration for racing.

Summary Table:

Case	Sprocket Ratio	Theoretical Top Speed	Practical Top Speed	Use Case
High	4.0 (15T/60T)	~125 km/h	Yes	Fast acceleration, tight tracks
Medium	3.0 (15T/45T)	~165 km/h	Yes (~150 km/h)	Balanced, general-purpose performance
Low	2.0 (15T/30T)	~250 km/h	No (too low torque)	Poor acceleration, not recommendec



Extra Calculations

Torque Curve Analysis using the given and calculated values.

Given:

- **Max engine power = 24 kW (restricted)**
- **Max engine speed = 10,000 rpm**
- **Torque formula:**

$$\text{Torque}(T) = \frac{Power \times 9550}{RPM}$$

Calculating Torque at Max RPM:

$$T = \frac{24000 \times 9550}{10000} = 22.92 \text{ Nm}$$

Assumed Simplified Torque Curve Shape:

Torque rises linearly from 0 Nm at 3,000 rpm (idle) to peak torque at 9,000 rpm, then drops slightly at max RPM (10,000 rpm).

Estimated Torque Values at Key RPMs:

Engine RPM	Torque (Nm)	Notes
3000	0	Assumed zero at idle
6000	20	Approximate linear rise
9000	35	Estimated peak torque
10000	22.9	Torque drops at max RPM

This simplified torque curve suits calculations for gear selection, acceleration, and drivetrain analysis.

Drivetrain Losses Calculation.

Assuming drivetrain efficiency $\eta = 90\%$ (0.9).

For example, at peak engine torque:

- **Engine torque at peak RPM:**

$$T_{engine} = 22.92 \text{ Nm(from previous calculation at max RPM)}$$

- **Overall gear ratio for 1st gear:**

$$\text{Gear ratio} \approx 2.85$$

- **Wheel torque:**

$$T_{wheel} = T_{engine} \times \text{Gear ratio} \times \eta$$
$$T_{wheel} = 22.92 \times 2.85 \times 0.9 \approx 58.7 \text{ Nm}$$

Summary:

- **Power and torque transmitted to the wheels are approximately 58.7 Nm at peak torque, after accounting for drivetrain losses (assuming 90%).**

Gear Ratios & Final Drive Calculation.

Given:

- **Example gear ratios (from a typical 6-speed KTM 390 gearbox):**
  - **1st Gear = 2.85**
  - **2nd Gear = 1.95**
  - **3rd Gear = 1.44**
  - **4th Gear = 1.16**

- 5th Gear = 0.96
- 6th Gear = 0.82

- Final drive sprocket ratio = 3 (from earlier sprocket calculation)

Overall Drive Ratios per Gear (Gear Ratio × Final Drive Ratio):

Gear	Gear Ratio	Overall Ratio (×3)
1st	2.85	8.55
2nd	1.95	5.85
3rd	1.44	4.32
4th	1.16	3.48
5th	0.96	2.88
6th	0.82	2.46

Acceleration & Performance Calculation.

Given:

- Wheel Torque (Nm): Calculated from engine torque, gear ratio, and drivetrain efficiency
- Wheel Radius: 0.264 m
- Vehicle mass (including driver): 290 kg (from earlier data)

Step 1: Calculate Wheel Torque

Using the earlier drivetrain loss calculation (~90% efficiency) and engine torque (~22.92 Nm at max RPM):

$$T_{wheel} = T_{engine} \times \text{Overall Gear Ratio} \times \eta$$

Gear	Overall Gear Ratio	Wheel Torque (Nm)
1st	8.55	22.92 × 8.55 × 0.9 ≈ 177.4
2nd	5.85	22.92 × 5.85 × 0.9 ≈ 120.5
3rd	4.32	22.92 × 4.32 × 0.9 ≈ 89.2
4th	3.48	22.92 × 3.48 × 0.9 ≈ 71.8
5th	2.88	22.92 × 2.88 × 0.9 ≈ 59.4

Step 2: Calculate Tractive Force (N)

$$F = \frac{T_{wheel}}{r}$$

Gear	Wheel Torque (Nm)	Tractive Force (N)
1st	177.4	177.4 / 0.264 ≈ 672.0
2nd	120.5	120.5 / 0.264 ≈ 456.1
3rd	89.2	89.2 / 0.264 ≈ 338.0
4th	71.8	71.8 / 0.264 ≈ 272.0
5th	59.4	59.4 / 0.264 ≈ 225.0

Step 3: Calculate Acceleration (m/s^2)

$$a = \frac{F}{m}$$

Gear	Tractive Force (N)	Acceleration (m/s^2)
1st	672.0	672.0 / 290 ≈ 2.32
2nd	456.1	456.1 / 290 ≈ 1.57
3rd	338.0	338.0 / 290 ≈ 1.17
4th	272.0	272.0 / 290 ≈ 0.94
5th	225.0	225.0 / 290 ≈ 0.78

Summary:

Gear	Wheel Torque (Nm)	Tractive Force (N)	Acceleration (m/s^2)
1st	177.4	672.0	2.32
2nd	120.5	456.1	1.57
3rd	89.2	338.0	1.17
4th	71.8	272.0	0.94
5th	59.4	225.0	0.78

This shows the theoretical maximum acceleration in each gear based on the KTM 390 engine specs, gear ratios, drivetrain efficiency, and wheel radius.

Fuel Efficiency & Energy Balance Refinement.

Given:

- **Engine power output (restricted):** $P_{engine} = 24$ **kW**
- **Powertrain efficiency:** $\eta_{pt} = 90\% = 0.90$ **(accounts for drivetrain losses)**
- **BSFC (Brake Specific Fuel Consumption) = 300 g/kWh = 0.3 kg/kWh (improved/race tuned value typical for KTM 390 racing)**
- **Fuel energy density = 44 MJ/kg**

Step 1: Calculate Power at Wheels

$$P_{wheel} = P_{engine} \times \eta_{pt} = 24 \times 0.9 = 21.6 \text{ kW}$$

Step 2: Fuel Consumption Rate at Wheel Power

$$\text{Fuel consumption rate } = BSFC \times P_{wheel} = 0.3 \times 21.6 = 6.48 \text{ kg/h}$$

Step 3: Energy Input from Fuel

Fuel energy input per hour:

$$E_{fuel} = \text{Fuel consumption} \times \text{Fuel energy density} = 6.48 \times 44 = 285.12 \text{ MJ/h}$$

Summary:

Parameter	Value	Unit
Power at wheels	21.6	kW
Fuel consumption rate	6.48	kg/h
Fuel energy input	285.12	MJ/h

This reflects the effective fuel quantity and energy required to overcome drivetrain losses, based on your KTM 390 engine's restricted power and realistic BSFC.

Next Step(Analysis)

1. **Dynamic simulation integration: Simulate lap times using these powertrain parameters with vehicle dynamics models.**
2. **Thermal analysis: Evaluate cooling requirements for engine, drivetrain based on power and losses.**
3. **Component stress analysis: Structural checks on drivetrain components using torque and loads calculated.**
4. **Optimization studies: Optimize gear ratios, fuel efficiency, and acceleration for specific tracks or performance goals.**